PROBLEM5:

PROBLEM STATEMENT:

Multi-Plan Telecommunications Cost Optimization: Evaluate several mobile plans against usage, validity, and budget constraints; calculate excess charges and identify the most economical eligible plan

OBJECTIVE:

To develop a Python program that compares multiple mobile plans based on data usage, call usage, validity, and budget constraints. The program calculates excess usage charges, determines the total cost of each plan, and identifies the most economical eligible plan.

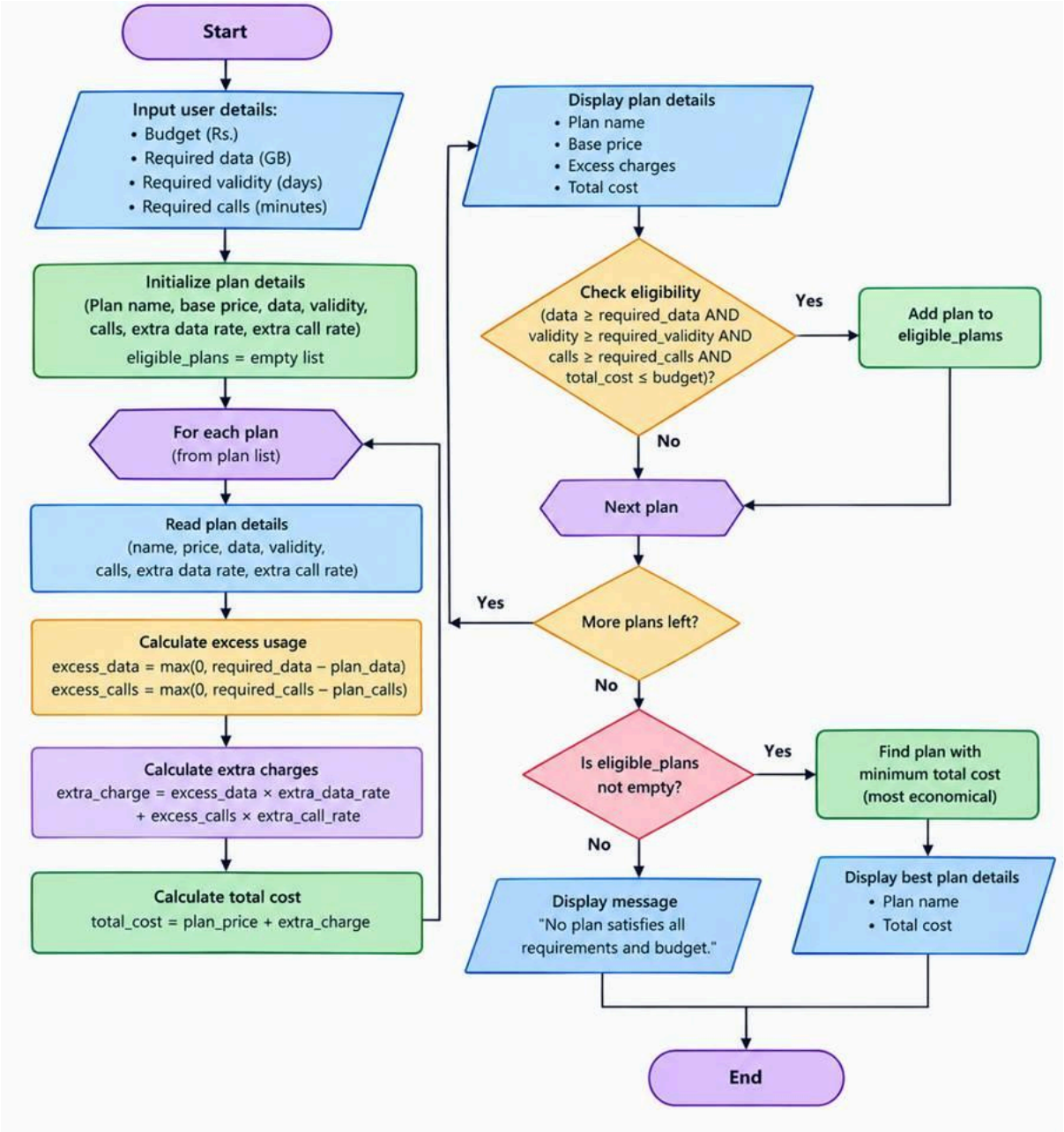
PROGRAM:

```
# Multi-Plan Telecommunications Cost Optimizationbudget = float(input("Enter your budget (Rs.): "))required_data = float(input("Enter required data (GB): "))required_validity = int(input("Enter required validity (days): "))required_calls = int(input("Enter required calls (minutes): "))plans = [ ["Plan A", 199, 1.5, 28, 300, 50, 0.50],          ["Plan B", 299, 3.0, 56, 600, 100, 0.30],          ["Plan C", 399, 5.0, 84, 1000, 150, 0.20]]eligible_plans = []print("\n--- Mobile Plan Comparison ---")for plan in plans:          name, price, data, validity, calls, extra_data, extra_call = plan    excess_data = max(0, required_data - data)    excess_calls = max(0, required_calls - calls)extra_charge = (excess_data * extra_data + excess_calls * extra_call)    total_cost = price + extra_charge    print("\nPlan:", name)    print("Base Price: Rs.", price)print("Excess Charges: Rs.", round(extra_charge, 2))    print("Total Cost: Rs.", round(total_cost, 2))  if (data >= required_data and            validity >= required_validity and              calls >= required_calls and
```

ALGORITHM:

1. Start.
2. Input the user's budget, required data, validity, and call minutes.
3. Store the details of multiple mobile plans.
4. For each plan, calculate excess data and call usage.
5. Calculate excess usage charges.
6. Calculate the total cost by adding the base price and excess charges.
7. Check whether the plan satisfies the data, validity, call, and budget requirements.
8. Add eligible plans to the eligible list.
9. Find the eligible plan with the lowest total cost.
10. Display the most economical eligible plan or a message if no plan qualifies.
11. Stop.

Flowchart



PSEUDOCODE:

BEGIN

INPUT budget, required_data

INPUT required_validity, required_calls

STORE mobile plan details

eligible_plans = empty list

```
        FOR each plan
        excess_data = MAX(0, required_data - plan_data)
        excess_calls = MAX(0, required_calls - plan_calls)

        extra_charge = excess_data * data_rate
                + excess_calls * call_rate

        total_cost = plan_price + extra_charge

        DISPLAY plan name, extra_charge, total_cost

        IF plan_data >= required_data AND
       plan_validity >= required_validity AND
        plan_calls >= required_calls AND
        total_cost <= budget THEN

        ADD plan to eligible_plans
        END IF
        END FOR

        IF eligible_plans is not empty THEN
        best_plan = eligible plan with minimum total_cost
        DISPLAY best_plan
        ELSE
        DISPLAY "No eligible plan found"
        END IF
END
```

OUTPUT:

Example input: Budget ₹400, required data 2 GB, validity 28 days, and calls 400 minutes.

--- Mobile Plan Comparison ---

Plan: Plan A

Base Price: Rs. 199

Excess Charges: Rs. 50.0

Total Cost: Rs. 249.0

Plan: Plan B

Base Price: Rs. 299

Excess Charges: Rs. 0.0

Total Cost: Rs. 299.0

Plan: Plan C

Base Price: Rs. 399

Excess Charges: Rs. 0.0

Total Cost: Rs. 399.0

Most Economical Eligible Plan:

Plan: Plan B

Total Cost: Rs. 299.0